

Assign 1: Linked List

If no LinkedList instance has been created yet, write statements in the main program to create a mylist instance with the ordered inputs as **the last 6 digits of your student's id** produce the output that **sort from small to large** (ascending order) by using the LinkedList class and Node class we have learned.

For Example: if student's id is 6706021 632032

then the last 6 digits are 632032

So the ordered inputs are 6, 3, 2, 0, 3, 2

Then we think what the outputs should be 0, 2, 2, 3, 3, 6

Note that: No problem with the input 0 and

No problem with the duplicate 2, 2, 3, 3. There is no restrict.

So we can write statements in main program like these:

mylist = LinkedList()

mylist.add(6)

mylist.add(3)

mylist.add(2)

mylist.add(0) # then we've the result 0, 2, 3, 6 but?

The next input is 3. Can we use method add() again?

Then, swap node position 1 and 6 by changing the node pointers using the LinkedList and Node classes. Do not use any methods that have not been taught in class. Do not change data in node. Modify pointers only, do not swap data.

The final output should be = ? 6, 2, 2, 3, 3, 0